SkillUp

Glossary – Generative AI: Elevate your Business Intelligence (BI) Career

Term	Definition
Akkio	A no-code AI platform allows users to create and deploy predictive models and generate reports.
Algorithmic bias	Bias occurring when machine learning models produce biased results due to biased training data or algorithm design.
Alteryx	AI tool used to connect to various data sources, blend the data, and apply cleansing operations such as removing duplicates and correcting errors, ensuring consistency and accuracy without manual intervention.
Anomaly detection	The process of identifying observations, events, or data points that deviate from the usual, standard, or expected behavior within a data set.
Apache Kafka	A distributed streaming platform capable of handling a high volume of real-time data.
Ask Data	A feature in Tableau that allows asking questions from data using natural language. Instead of manually dragging and dropping fields, type a question in common language, and Ask Data instantly responds with automatic data visualizations.
Automated dashboard creation	A process wherein once the AI tools have the data, they create a complete dashboard with related charts and visualizations.
Automated learning	The process of automatically learning and capturing the important features of your data.
Automatic data visualization	A process whereby AI tools analyze data and recommend the most appropriate chart type to present insights effectively.
Automatic schema understanding	Refers to comprehending and designing the structure, organization, and relationships of tables within a data set.
Azure Cognitive Services	A collection of cloud-based APIs by Microsoft that use artificial intelligence (AI) to enhance user experiences.
Bias	Systematic errors that lead to inaccurate, prejudiced, or unrepresentative insights.
Business intelligence (BI) lifecycle	An iterative process that transforms data into actionable insights.
ChatGPT	OpenAI can help BI analysts craft compelling narratives based on data insights, draft initial reports, and write prompts for data analysis.
Clustering	Unsupervised machine learning task of grouping similar data points together based on their characteristics or attributes to form groups.
Code completion	Refers to when AI suggests relevant SQL keywords and functions as the script is written.
Confirmation bias	Bias referring to analysts interpreting data to confirm existing beliefs leading to the selective use of data that supports a desired outcome.
Comprehensive ethical AI frameworks	Sets of guidelines and principles designed to address ethical concerns such as fairness, accountability, transparency, privacy, and safety.
Copilot	Microsoft tool that integrates with Microsoft products and helps BI professionals through its natural language processing capabilities in data analysis, data visualization, and reporting.
Correlation analysis	A statistical method used to determine if there is a relationship between two variables and how strong the relationship is.
Cost-based optimization	The technique of database query optimization to ensure the effective execution of the SQL script and reduce query execution time.
Data analysis	The process of collecting, modeling, and analyzing data using various statistical and logical methods and techniques.
Data cleansing and preparation	The process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a data set.
Data collection	Systematic process of gathering and recording information from various sources for analysis, interpretation, and decision making.
Data lineage	A process of tracking the flow of data over time, providing a clear understanding of where the data originated, how it has changed, and its ultimate destination within the data pipeline.
Data mapping	Refers to when AI tools provide a data point overview.
Data profiling	Process of reviewing and analyzing data sets to understand their structure and information.
Data reporting	Systematic process of presenting information in a meaningful and accessible format.
Data type inference	Refers to when the AI tool automatically infers the data type of columns within tables. This inference ensures proper filtering and aggregation functions used in the generated SQL script.
Data visualization	Graphical representation of information and data using visual elements such as charts, graphs, and maps.
Database schema analysis	A process where AI helps querying structures to estimate the execution cost of different query plans.
DataRobot	AI tool that analyzes the data characteristics, evaluates the different clustering models available, and chooses the model most suitable for the data.
Dummy customer	Mimic real customer behavior, but not a copy of any real person, thereby safeguarding customer privacy.
Explainable AI	Techniques and processes that make decisions of AI models understandable to humans.
Federated learning approach	A system by which models across decentralized data sources are trained and updates shared, enhancing privacy and diversity in training data.
Gamma.app	An AI-powered platform designed for creating various types of content, including presentations, documents, and websites.
Gemini	Google tool for data storytelling with conversational abilities that could be leveraged for brainstorming and generating ideas.
Generative adversarial network (GAN)	Type of generative model that includes two neural networks: generator and discriminator. The generator is trained on vast data sets to create samples like text and images. The discriminator tries to distinguish whether the sample is real or fake.
Generative artificial intelligence (GenAI)	Type of AI that can generate new content like text, images, music, audio, and videos using generative models. Generative AI models can learn patterns and structures from their input training data and then generate new data with similar characteristics.
Gretel.ai	A synthetic data platform that creates synthetic versions of existing data in minutes and can anonymize and enrich data, enabling businesses to generate valuable insights while ensuring privacy.
H2O.ai	Software company that provides open-source and proprietary tools for machine learning and AI.
Hazy	A generative AI-based tool that helps businesses generate synthetic data that can be safely used for testing, analytics, and AI training without risking sensitive information.
IBM Cognos Analytics	Web-based intelligence suite powered by IBM watsonx, it offers an embedded assistant that allows users to input text to generate insights into the data.
IBM Watson Discovery	Intelligent document processing engine that helps to gain insights from complex business documents.
Image recognition	A software' s ability to identify and classify people, objects, places, writing, and actions in digital images and video.
Imputing data	BI technique that involves filling in missing values with estimated values.
Index recommendation	Process wherein AI suggests creating indexes on frequently used columns to speed up data retrieval.
Infogram	An infographics platform that leverages AI to revolutionize data visualization and sharing. With Infogram, you can create interactive charts, maps, and infographics.
Intelligent auto-completion	Refers to when AI tools suggest relevant keywords, functions, and even additional expressions as you write SQL code.
Interactive data visualization	Method of presenting complex data sets in an engaging and comprehensible manner wherein of static charts or tables, it employs dynamic visuals that allow users to interact with the information, drilling down into specifics or zooming out for broader perspectives.
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Joining optimization	A technique wherein AI identifies the optimal join order for complex queries with multiple tables.
Large language models (LLMs)	A type of deep learning model trained on massive amounts of text data to learn the patterns and structures of language. They can perform language-related tasks, including text generation, translation, summarization, sentiment analysis, and more.
Machine learning	Type of artificial intelligence that focuses on creating algorithms and models that enable computers to learn and make predictions or decisions.
Measurement bias	Bias arising from inaccuracies in data collection, such as flawed survey designs, incorrect data entries, or biased questions that influence responses.
Misrepresentation	False representation of data.
missing at random (MAR)	Refers to missing values that are not random but dependent on certain factors.
missing completely at random (MCAR)	Refers to missing values unrelated to other variables where the occurrence is entirely by chance.
missing not at random (MNAR)	Refers to missing values related to the value itself.
Mostly AI	Generative AI-based tool that can generate synthetic data.
Multidimensional Baselining	The process of establishing normal behavior patterns across multiple data points or variables.
Narrative Science	A company that uses automated narrative generation for storytelling from data.
Natural language generation	The use of artificial intelligence to produce spoken or written human-like text.
Natural language processing (NLP)	Branch of artificial intelligence that enables computers to understand, manipulate, and generate human language (natural language).
Natural language querying	A capability in BI software that enables users to ask questions of data in natural language.
NLP engine	Technology that is used by machines to understand, process, and generate human language.
NoSQL	Refers to nonrelational databases that use a nontabular format to store data. Unlike traditional relational databases, which rely on structured tables, NoSQL databases offer more flexible data models.
Otter.ai	Web-based transcription service that uses AI to transcribe conversations, take notes during meetings, and even attend meetings as a substitute.
Outerbase	A freely available AI-powered interface that can be used to manage your database.
Overfitting in machine learning	When a model gives accurate predictions for training data but not for new data. This happens when the model learns the details and noise of the training data and can' t generalize to new examples.
Picture classification	Assigning a label or tag to an entire image based on already labeled images of training data.
Power BI	BI tool developed by Microsoft that helps transform raw data from various sources into meaningful insights through interactive visualizations and reports.
Predictive analytics	Statistical techniques used to predict future behavior and events by uncovering relationships and patterns within large volumes of data.
Qlik Sense	BI tool that integrates advanced analytics, enabling users to perform sentiment analysis and predictive modeling within their dashboards.
Scenario modeling	Refers to when generative AI helps formulate SQL queries to explore hypothetical scenarios on different parameters.
Segmentation	The breaking down of large data sets into smaller, more manageable parts.
Selection bias	Bias occurring when the data sample is not representative of the intended population.
Semi-supervised learning	A process where AI tools combine labeled and unlabeled data to identify patterns in the labeled data that serve as a guide to assign labels to the unlabeled data.
Sentiment analysis	The process of analyzing large volumes of text to determine whether it expresses a positive, negative, or neutral sentiment.
Shapley Additive exPlanations (SHAP)	Game-theoretic approach used to explain the output of any machine learning model, which connects optimal credit allocation with local explanations by leveraging classic Shapley values from game theory and their related extensions.
Speech synthesis	Computer-based generation of artificial speech from normal language text.
SQL generation	Refers to the process of creating SQL (structured query language) queries automatically, without manually writing the code.
Structured query language (SQL)	Standardized programming language used to interact with relational database management systems (RDBMS) which enables you to store, manipulate, and retrieve data from relational databases.
Tableau	Visual analytics platform that transforms the way we use data to solve problems.
text2sql.ai	An AI-powered tool that generates SQL queries from natural language instructions.
Transformers	A deep learning architecture that uses an encoder-decoder mechanism. Transformers can generate coherent and contextually relevant text.
Variational autoencoder (VAE)	A type of generative model that is basically a neural network model designed to learn the efficient representation of input data by encoding it into a smaller space and decoding back to the original space.